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Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity

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Abstract

Superconducting waveguides are a promising platform for ultralow-loss transmission in the millimeter-wave to terahertz band under cryogenic conditions, with potential applications in astronomical instrumentation and emerging quantum technologies. We develop a framework, based on microscopic superconductivity theory, to evaluate the power-flow attenuation constant α of superconducting rectangular waveguides in the 100 GHz–THz range, applicable to arbitrary electronic mean free paths ℓ from the dirty limit $\ell \ll \xi_0$ to the clean limit $\ell \gg \xi_0$. We also derive an analytical expression for two-level-system (TLS)-induced attenuation α_{TLS} in thin native oxide layers within the standard TLS model. Using this framework, we perform numerical evaluations of α for representative materials over standard waveguide sizes from WR15 to WR1. In the high-frequency regime $f \gtrsim 0.5\Delta/h$, low attenuation favors the clean regime $\ell \gtrsim \xi_0$, indicating that high-purity materials can achieve very low attenuation below their gap frequency. For the TLS contribution, using parameter values representative of native Nb oxides, we find that α_{TLS} can become relevant at sufficiently low temperatures $T/T_c \lesssim 0.1\text{--}0.2$, where quasi-particle dissipation is exponentially suppressed. Finally, we extend the discussion to the strong-excitation regime using a recently developed nonlinear-response theory within the Keldysh–Usadel framework of nonequilibrium superconductivity and show that nonlinear dissipation produces a Higgs-mode peak in α near $f \simeq \Delta/h$ via a Kerr-type nonlinearity of the dissipative conductivity. This peak provides a distinct hallmark of the Higgs mode that has been largely overlooked so far.

1. Introduction

The millimeter–submillimeter region bridges the microwave and infrared regimes and remains a challenging yet important part of the spectrum [1–11]. Superconducting waveguides in this band are attracting growing attention across multiple frontiers of science and technology.

In quantum technologies, recent efforts to push superconducting quantum hardware toward higher operating frequencies [2, 3] seek to relax refrigeration requirements by reducing thermal occupation, which can enable operation at elevated temperatures. In this context, low-loss interconnects extending from the microwave into the millimeter-wave regime, and potentially toward the sub-terahertz range, are expected to become increasingly important. As system scaling increases routing distances, distributed attenuation becomes a key figure of merit.

In astronomical instrumentation operating in the 100 GHz to THz range [1, 4, 5], key observational windows include the cosmic microwave background [6], molecular rotational lines and dust continuum emission [7, 8]. Millimeter- and submillimeter-wave bands are also being explored for technosignature

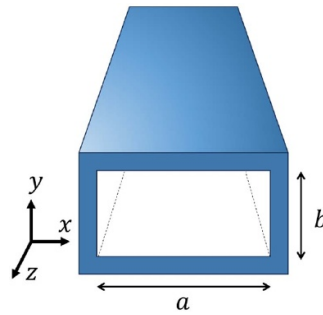


Figure 1. Schematic of a superconducting rectangular waveguide. The wall thickness is assumed to be much larger than the electromagnetic-field penetration depth of the superconducting walls.

searches in the search for extraterrestrial intelligence [9–11]. In such cryogenic front ends, the delivered signal power can be extremely small, which places stringent constraints on transmission loss along waveguide runs.

Normal-conducting waveguides have long been central to millimeter- and submillimeter-wave systems [4, 5], offering relatively low attenuation [12] owing to their hollow metallic geometry. As the frequency is pushed toward the terahertz regime, however, conductor loss increases and low-loss propagation becomes a key bottleneck. One promising route to mitigating such attenuation is the use of superconducting waveguides (figure 1). Kurpiers *et al* [13] measured attenuation in commercially available microwave-frequency waveguides at cryogenic temperatures. More recently, Nakajima *et al* [14] fabricated a rectangular waveguide from bulk niobium and demonstrated an attenuation of $5 \times 10^{-4} \text{ dB cm}^{-1}$ at a frequency of $f \simeq 100 \text{ GHz}$ and at a temperature of $T \simeq 4\text{--}5 \text{ K}$, which is substantially smaller than that of comparable normal-metal waveguides. These results highlight the potential of superconducting waveguides for ultralow-loss transmission at millimeter-wave frequencies and, potentially, toward the terahertz regime.

For weakly lossy waveguides, provided the operating frequency point is sufficiently above cutoff and sufficiently below the superconducting transition temperature T_c , the attenuation can be obtained to first order by treating the wall surface impedance as a perturbation to the fields of the corresponding lossless waveguide solution [15, 16]. In this regime, the attenuation dominated by thermally excited quasiparticles is directly related to the superconducting surface resistance R_s . The central theoretical task therefore reduces to calculating R_s . Despite this conceptual simplicity, a systematic theoretical treatment of attenuation (α) in superconducting waveguides has received relatively limited attention. Only a small number of studies (see, e.g. [16] and references therein) have addressed attenuation from the perspective of microscopic superconductivity theory. Moreover, existing treatments rely on the dirty-limit Mattis–Bardeen formalism for the linear-response complex conductivity, although practical materials are not necessarily in that limit.

This gap in the literature is not limited to the treatment of superconducting conductor loss. At sufficiently low temperatures, quasiparticle-induced dissipation becomes exponentially small, and dielectric loss associated with the so-called tunneling two-level systems (TLS) [17–19] in the thin native oxide layer covering the superconducting surface can begin to dominate. To our knowledge, the corresponding TLS-induced attenuation in superconducting waveguides has not yet been evaluated in a systematic manner. Such an assessment is likely to become increasingly important if superconducting waveguides are deployed at very low temperatures, for example in the millikelvin regime.

In this paper, we evaluate the attenuation constant α of superconducting waveguides, with a deliberate focus on those with *rectangular* geometries. The motivation is that, in astronomical instrumentation, waveguides are increasingly required to provide not only low-loss transmission but also circuit functionality such as filtering, channelization, and multiplexing [4, 5]. In the millimeter- and submillimeter-wave bands, such functional waveguide circuits are routinely realized as compact computer numerical control-milled metal blocks based on rectangular-waveguide technology, which offers mature design methodologies and straightforward mechanical integration. In quantum hardware, similar requirements are not yet as widely established, but they are likely to emerge as operating frequencies move into the millimeter-wave regime and system scaling increases routing distances, making frequency-selective routing and multiplexing increasingly attractive. This prospect further strengthens the case for rectangular waveguides as a practical platform.

We evaluate α in the 100 GHz–THz range using a microscopic theory of superconductivity that is applicable to arbitrary electronic mean free paths ℓ , spanning the dirty limit $\ell \ll \xi_0$ to the clean limit $\ell \gg \xi_0$. Specifically, we compute the linear-response complex conductivity using the Eilenberger formalism (see, e.g. [20–22] and references therein), which remains valid across this wide range of ℓ . This yields a general framework for estimating waveguide attenuation across a broad parameter space. We also investigate dielectric loss due to TLS defects, which can become relevant at sufficiently low temperatures where quasiparticle dissipation is strongly suppressed. Owing to the simple analytic form of the electric-field distribution in a rectangular waveguide, the TLS-induced loss and the resulting attenuation can be expressed in a compact analytical form involving complete elliptic integrals of the first and second kinds.

In addition to the weak-signal limit, we investigate power-dependent attenuation under strong excitation based on the nonlinear response theory of superconductivity, which becomes relevant when high-power signals are transmitted. Here we focus on the dirty-limit regime, in which a microscopic formulation of nonlinear dissipation has recently been advanced within the Keldysh–Usadel framework of nonequilibrium superconductivity [23]. Beyond this practical motivation, the nonlinear regime is also of fundamental interest because it provides a natural setting for the emergence of the Higgs mode. The Higgs mode is a collective amplitude excitation of the superconducting order parameter [24] and has attracted sustained interest in condensed-matter physics [23–39].

The paper is organized as follows. In section 2, we establish a theoretical framework for evaluating attenuation α in superconducting rectangular waveguides, applicable across the entire range of mean free paths ℓ (from $\ell \ll \xi_0$ to $\ell \gg \xi_0$) and for arbitrary T and ω . In the same section, we also derive an analytical expression for TLS-induced attenuation α_{TLS} . In section 3, we apply the framework to representative materials and compute the resulting attenuation over frequencies ranging from several tens of gigahertz to the terahertz regime. In section 4, we turn to the strong-excitation regime and analyze attenuation within a nonlinear framework based on a recent microscopic nonequilibrium theory [23]. Finally, section 5 summarizes the main conclusions and discusses future directions.

2. Theory

In this section, we develop a microscopic framework for evaluating the power-flow attenuation α in superconducting rectangular waveguides within linear response, applicable to arbitrary electronic mean free paths ℓ , and we also derive an analytical expression for the TLS-induced attenuation α_{TLS} .

2.1. BCS superconductors with nonmagnetic impurity scattering

We begin with a brief review of BCS superconductors with nonmagnetic impurity scattering, characterized by a mean free path ℓ ranging from the clean limit ($\ell \gg \xi_0$) to the dirty limit ($\ell \ll \xi_0$) (see, e.g. [40]). Here $\xi_0 = \hbar v_f / \pi \Delta_0$ is the BCS coherence length, with v_f the Fermi velocity and Δ_0 the superconducting energy gap at zero temperature. In the following we focus on the extreme type-II regime where the London penetration depth is much larger than the coherence length ($\lambda \gg \xi$), which ensures that the local formulation of the conductivity and surface impedance is applicable.

In the zero-current limit, as a consequence of Anderson’s theorem, the equilibrium solution of the Eilenberger equation does not depend on the mean free path. The normal and anomalous Matsubara Green’s functions are given by $g_m = \hbar\omega_m / \sqrt{(\hbar\omega_m)^2 + \Delta^2}$ and $f_m = \Delta / \sqrt{(\hbar\omega_m)^2 + \Delta^2}$, respectively. Here, $\hbar\omega_m = 2\pi k_B T(m + 1/2)$ is the Matsubara frequency, and Δ is the pair potential at the temperature T . The pair potential is determined self-consistently through the gap equation

$$\ln \frac{T_c}{T} = 2\pi k_B T \sum_{\omega_m > 0} \left(\frac{1}{\hbar\omega_m} - \frac{f_m}{\Delta} \right), \quad (1)$$

where T_c is the superconducting critical temperature. Using this $\Delta(T)$, the penetration depth $\lambda(\ell, T)$ can be calculated (see, e.g. [21, 40] and references therein) as

$$\frac{1}{\lambda^2(\ell, T)} = \frac{2\pi k_B T}{\lambda_0^2} \sum_{\omega_m > 0} \frac{f_m^2}{d_m}, \quad (2)$$

where $\lambda_0^{-2} := \lambda^{-2}(0, 0) = (2/3)\mu_0 e^2 N_0 v_f^2 = \mu_0 e^2 n / m$ is the BCS penetration depth in the clean limit at zero temperature, and $d_m := \gamma + \sqrt{(\hbar\omega_m)^2 + \Delta^2}$. Here N_0 is the normal-state density of states at the Fermi energy, and γ is the impurity scattering rate, which is related to the mean free path ℓ through $\gamma / \Delta_0 = \pi \xi_0 / 2\ell$.

In the dirty limit ($\ell \ll \xi_0$ or $\gamma/\Delta_0 = \pi \xi_0/2\ell \gg 1$), substituting $d_m \simeq \gamma$ into equation (2) yields the dirty-limit expression, $\lambda^{-2}(\ell \ll \xi_0, T) = (\Delta/\Delta_0) \tanh(\Delta/2k_B T) \lambda_{0,\text{dirty}}^{-2}$. Here, the constant $\lambda_{0,\text{dirty}}^{-2} = (\pi \Delta_0/2\gamma) \lambda_0^{-2} = \pi \mu_0 \Delta_0 \sigma_n/\hbar$ is the well-known BCS penetration depth in the dirty limit at $T \rightarrow 0$. Although this dirty-limit formula is widely employed in the analysis of experimental data on superconducting devices, its range of validity is restricted. Therefore, to evaluate λ for arbitrary (ℓ, T) , one must in principle compute equation (2) numerically.

2.2. Complex conductivity $\sigma(\ell, T, \omega)$

The complex conductivity $\sigma = \sigma_1 + i\sigma_2$, including impurity scattering effects, can be derived as the linear response to an ac electromagnetic field within the Keldysh–Eilenberger theory of nonequilibrium superconductivity. This yields a well-established general expression in terms of real-frequency Green's functions [20, 21] (see also [37] for the case in which a dc current is superposed on the ac field). Accordingly, $\sigma(\ell, T, \omega)$ is given by [20–22]

$$\sigma(\ell, T, \omega) = \sigma_1 + i\sigma_2 = \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)], \quad (3)$$

$$\kappa = \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-}, \quad (4)$$

$$\kappa^a = \frac{1}{3} \frac{-g_+ g_-^* - f_+ f_-^* - 1}{d_+ - d_-^*}. \quad (5)$$

In equations (3)–(5), the normal-state dc conductivity is $\sigma_n = (2/3)e^2 N_0 v_f^2 \tau$, where $\tau = \ell/v_f$ is the electron relaxation time; we define $\epsilon_{\pm} := \epsilon \pm \hbar\omega/2$ and set $g_{\pm} := g(\epsilon_{\pm})$, $f_{\pm} := f(\epsilon_{\pm})$, $d_{\pm} := d(\epsilon_{\pm})$, and $\mathcal{T}_{\pm} := \mathcal{T}(\epsilon_{\pm})$, where $g(\epsilon) = -i(\epsilon + i0)/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $f(\epsilon) = -i\Delta/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $d(\epsilon) = i\gamma + i\sqrt{\Delta^2 - (\epsilon + i0)^2}$, and $\mathcal{T}(\epsilon) := \tanh(\epsilon/2k_B T) = 1 - 2f_{\text{FD}}(\epsilon)$, with $f_{\text{FD}}(\epsilon)$ the Fermi–Dirac distribution. In the normal state ($\Delta \rightarrow 0$), we have $g_{\pm} \rightarrow 1$, $f_{\pm} \rightarrow 0$, and $d_+ - d_-^* = \hbar\omega + 2i\gamma$. Equation (3) then reduces to the Drude form of the ac conductivity in normal metals: $\sigma_1 \rightarrow \sigma_n/[1 + (\omega\tau)^2]$ and $\sigma_2 \rightarrow \sigma_n \omega\tau/[1 + (\omega\tau)^2]$.

In the dirty limit ($\ell \ll \xi_0$, equivalently $\gamma/\Delta_0 = \pi \xi_0/2\ell_{\text{imp}} \gg 1$), we may approximate $d_+ \simeq d_- \simeq i\gamma$. Under this approximation, equation (3) reduces to the Mattis–Bardeen expressions for the complex conductivity in the dirty limit [41, 42].

$$\sigma_1(T, \omega) = \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon [f_{\text{FD}}(\epsilon) - f_{\text{FD}}(\epsilon')] M_1(\epsilon, \omega), \quad (6)$$

$$M_1(\epsilon, \omega) = \text{Reg}(\epsilon) \text{Reg}(\epsilon') + \text{Ref}(\epsilon) \text{Ref}(\epsilon'), \quad (7)$$

and

$$\sigma_2(T, \omega) = \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon \tanh \frac{\epsilon}{2k_B T} M_2(\epsilon, \omega), \quad (8)$$

$$M_2(\epsilon, \omega) = \text{Reg}(\epsilon) \text{Im}f(\epsilon') + \text{Ref}(\epsilon) \text{Im}f(\epsilon'). \quad (9)$$

Here, $\epsilon' := \epsilon + \hbar\omega$. It is clear that the widely used convenient formulas, equations (6) and (8), are valid only in the dirty limit. They are generally insufficient for a quantitative description of dissipation in realistic superconducting devices, which typically do not fall strictly into the dirty-limit regime. In the following, we therefore employ equation (3), which is applicable to superconductors with arbitrary mean free paths.

Figures 2(a) and (b) shows the real and imaginary parts of the complex conductivity for a clean superconductor ($\ell = 10\xi_0$), while figures 2(c) and (d) presents the corresponding results for a moderately dirty superconductor ($\ell = \xi_0$). These behaviors are well established in the classic literature (see, e.g. [20, 21, 43] and references therein). The plots shown here nevertheless provide a convenient reference for the analysis that follows, since much of the recent superconducting-device literature emphasizes the dirty limit, while the moderately clean and clean regimes are more rarely addressed.

It should be emphasized that, in the low-frequency limit ($\hbar\omega/\Delta_0 \rightarrow 0$), the integrand may be expanded in the small parameter ω . In this case, the imaginary part of equation (3) reduces to $(\sigma_n/2\tau\omega) \int d\epsilon \mathcal{T}(\epsilon) \text{Re}(f^2/d)$, which can be recast as a Matsubara-frequency sum, yielding (see, e.g. [21])

$$\sigma_2(\ell, T, \omega) \Big|_{\omega \rightarrow 0} = \frac{1}{\mu_0 \omega \lambda^2(\ell, T)}, \quad (10)$$

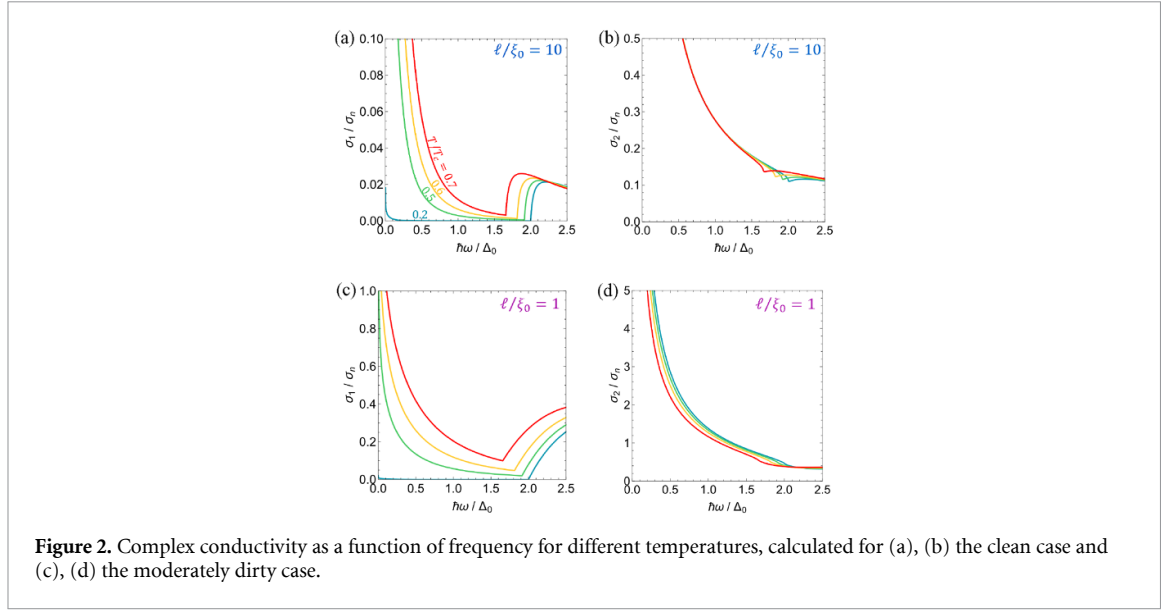


Figure 2. Complex conductivity as a function of frequency for different temperatures, calculated for (a), (b) the clean case and (c), (d) the moderately dirty case.

which corresponds to the well-known low-frequency relation between σ_2 and λ . This relation is valid only in the low-frequency, low-temperature regime and does not hold in many situations. In appendix A, $\sigma_2(\ell, T, \omega)$ is compared with $\sigma_{2,\text{low}} := 1/\mu_0\omega\lambda(\ell, T)$.

2.3. Surface resistance

Having derived the complex conductivity $\sigma = \sigma_1 + i\sigma_2$, we now express the surface impedance $Z_s = R_s - iX_s$ in terms of σ , where R_s and X_s denote the surface resistance and reactance, respectively. This step is a straightforward application of classical electrodynamics: the superconducting physics enters only through σ , and no additional microscopic detail is required.

In the low-frequency regime, $\hbar\omega \ll \Delta$, using the approximation $\sigma_1/\sigma_2 \ll 1$ together with the low-frequency relation $\sigma_2 \simeq \sigma_{2,\text{low}} = 1/(\mu_0\omega\lambda^2)$ (see equation (10) and appendix A), one obtains the well-known results: $R_s = (1/2)\mu_0^2\omega^2\lambda^3\sigma_1$ and $X_s = \sqrt{\mu_0\omega/\sigma_2}$. However, for our purposes it is necessary to evaluate Z_s under general conditions, where the simplifying assumptions $\sigma_1/\sigma_2 \ll 1$ and $\sigma_2 \simeq \sigma_{2,\text{low}}$ are not necessarily satisfied. It is therefore instructive to derive the general expression for Z_s in terms of σ_1 and σ_2 .

Starting from Maxwell's equations together with the relation $\mathbf{J} = \sigma\mathbf{E}$, where σ is the complex conductivity (equation (3)) and $\mathbf{E}$ is the ac electric field, one arrives at the well-known relation $\nabla^2(\mathbf{E}, \mathbf{H}) = (\mathbf{E}, \mathbf{H})/\Lambda^2$. Here, $\Lambda := (-i\mu_0\omega\sigma)^{-1/2}$, which can be expressed as

$$\Lambda(\ell, T, \omega) = \frac{1}{\sqrt{2\mu_0\omega}} \left(\frac{\sqrt{|\sigma| + \sigma_2}}{|\sigma|} + i \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|} \right). \quad (11)$$

It should be noted that under the assumption $\sigma_1/\sigma_2 \ll 1$, one finds $\Lambda \rightarrow \lambda$.

The solution of the one-dimensional equation $d^2E/d\eta^2 = E/\Lambda^2$, where η denotes the depth from the surface of a superconductor, is given by $E(\eta) = E_0 e^{-\eta/\Lambda}$. Substituting this into the definition of the surface impedance yields $Z_s(\ell, T, \omega) = E_0 / \int_0^\infty J(\eta) d\eta = \sqrt{-i\mu_0\omega/\sigma} = 1/\sigma\Lambda$. The real part corresponds to the surface resistance,

$$R_s(\ell, T, \omega) = \sqrt{\frac{\mu_0\omega}{2}} \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|}. \quad (12)$$

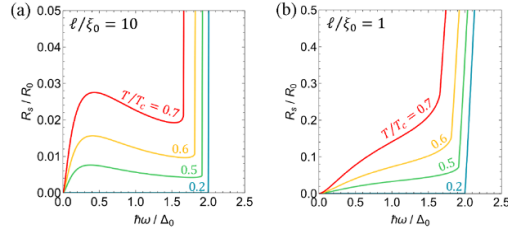
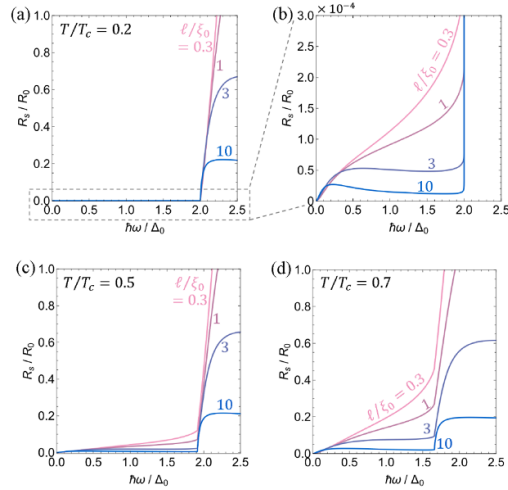
Using $|\sigma| - \sigma_2 \simeq \sigma_1^2/(2\sigma_2)$ and $\sigma_2 \simeq 1/(\mu_0\omega\lambda^2)$ for $\sigma_1/\sigma_2 \ll 1$, this reproduces the familiar low-frequency expression $R_s = (1/2)\mu_0^2\omega^2\lambda^3\sigma_1$. The imaginary part corresponds to the surface reactance, $X_s = \sqrt{\mu_0\omega/2} \sqrt{|\sigma| + \sigma_2}/|\sigma|$, which reduces to the low-frequency expression $X_s \simeq \sqrt{\mu_0\omega/\sigma_2}$ in the limit $\sigma_1/\sigma_2 \ll 1$.

It is useful to introduce a unit to express the surface resistance in a dimensionless form, which enables material-independent discussions of the surface resistance. For this purpose, we define a constant R_0 in terms of the zero-temperature, pure-limit material parameters as

$$R_0 := \frac{\mu_0\Delta_0\lambda_0}{\hbar}. \quad (13)$$

Table 1. Parameters required to compute the normalized complex conductivity σ/σ_n and the normalized surface resistance R_s/R_0 .

ℓ/ξ_0	Normalized electronic mean free path
T/T_c	Normalized temperature
$\hbar\omega/\Delta_0$	Normalized angular frequency of photon

**Figure 3.** Surface resistance as a function of frequency for different temperatures, calculated for (a) clean and (b) moderately dirty cases.**Figure 4.** Surface resistance as a function of frequency for different mean free paths, ranging from relatively dirty to relatively clean regimes, calculated for (a), (b) $T/T_c = 0.2$, (c) 0.5 , and (d) 0.7 .

Here, λ_0 denotes the zero-temperature London penetration depth in the pure limit. With this definition, the surface resistance can be written in terms of dimensionless parameters as

$$\frac{R_s(\ell, T, \omega)}{R_0} = \sqrt{\frac{\pi}{2}} \frac{1}{\sqrt{\ell/\xi_0}} \sqrt{\frac{\hbar\omega}{\Delta_0}} \sqrt{\frac{|\sigma| - \sigma_2}{\sigma_n}} \frac{\sigma_n}{|\sigma|}. \quad (14)$$

For the reader's convenience, the parameters required to evaluate equation (14) are summarized in table 1.

Figure 3 shows how the spectrum of the surface resistance evolves with temperature. For both the clean case ($\ell/\xi_0 = 10$) and the moderately dirty case ($\ell/\xi_0 = 1$), R_s increases with T as expected. Figure 4 shows the surface resistance R_s as a function of frequency for different mean free paths ℓ and temperatures T . Figure 4(a) shows the results for $T/T_c = 0.2$, where the surface resistance is strongly suppressed for $\hbar\omega < 2\Delta_0$ for all values of the mean free path, since only an exponentially small number of quasiparticles are thermally excited at this temperature. Figure 4(b) provides an enlarged view of figure 4(a), highlighting that the dependence of R_s on the mean free path becomes significant. Figures 4(c) and (d) show the corresponding results for higher temperatures, $T/T_c = 0.5$ and 0.7 , respectively, demonstrating that the structure observed in figure 4(b) becomes more pronounced as the temperature increases. Over nearly the entire frequency range, R_s increases as the mean free path decreases. An exception arises in the regime $\hbar\omega \ll \Delta_0$, where a dirty material does not necessarily yield a larger R_s ; indeed, R_s is minimized around $\ell/\xi_0 \sim 1$. Qualitatively, this follows from the low-frequency relation $R_s \propto \lambda^3 \sigma_1$: λ^3 and σ_1 are, respectively, decreasing and increasing functions of ℓ , and their competition

Table 2. Dimensions and frequency ranges of standard rectangular waveguides. Here a and b are the broad and narrow wall dimensions, respectively.

Waveguide name	a [mm]	b [mm]	Cutoff of TE ₁₀ [GHz]	Cutoff of next mode [GHz]
WR28	7.112	3.556	21.1	42.2
WR15	3.759	1.880	39.9	79.7
WR10	2.54	1.27	59.0	118.0
WR8	2.032	1.016	73.8	147.5
WR5.1	1.295	0.648	115.7	231.4
WR3.4	0.864	0.432	173.6	347.1
WR2	0.508	0.254	295.1	590.1
WR1	0.254	0.127	590.1	1180.3

produces the observed minimum. This behavior is well established theoretically (see e.g. [21, 44]), supported by consistent experimental trends (see, e.g. [45, 46]), and is widely recognized within the superconducting accelerator community.

2.4. Attenuation constant

We briefly summarize some standard textbook results for a perfectly conducting rectangular waveguide [15]. Consider a vacuum-filled lossless waveguide with cross section $a \times b$, where $x \in [0, a]$ and $y \in [0, b]$ (see figure 1). For the dominant TE₁₀ mode, assuming the fields are proportional to $e^{-i\omega t} e^{i\beta z}$, we have $H_x = H_0(\beta c^2 \pi / i \omega_c^2 a) \sin(\pi x/a)$, $H_y = 0$, $H_z = H_0 \cos(\pi x/a)$, $E_x = 0$, $E_y = E_0 \sin(\pi x/a)$, and $E_z = 0$, where $E_0 = i(\omega/\omega_c) c \mu_0 H_0$, $\beta = \sqrt{(\omega/c)^2 - (\pi/a)^2}$, $\omega_c = \pi c/a$ and c is the speed of light in vacuum. The time-averaged power flow along the z direction is

$$P_{\text{flow}} = \frac{1}{2} \int dx dy \text{Re}[(\mathbf{E} \times \mathbf{H}^*) \cdot \hat{\mathbf{z}}] = \frac{\mu_0 \omega \beta a^3 b}{4\pi^2} H_0^2 = \frac{\beta ab}{4\mu_0 \omega} |E_0|^2, \quad (15)$$

which is independent of z , meaning that the power flow does not attenuate in the lossless case. Here P_{flow} denotes the guided power crossing a transverse cross section and has units of power.

We now introduce a small wall loss perturbatively, starting from the lossless field distribution derived above. When a perturbative surface impedance $|Z_s| \ll Z_0$, where $Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 376.73 \, \Omega$, is introduced, the field distribution remains essentially unchanged, but the power flow attenuates due to surface losses:

$$P_{\text{loss}} = \frac{1}{2} R_s \oint d\ell |\mathbf{H}_{\parallel}|^2 = \frac{1}{2} R_s \langle H_{\parallel}^2 \rangle (2a + 2b), \quad (16)$$

where

$$\langle H_{\parallel}^2 \rangle := \frac{\oint d\ell |\mathbf{H}_{\parallel}|^2}{2a + 2b} = \frac{H_0^2}{2 + (2b/a)} \left(\frac{2b}{a} + \frac{\omega^2}{\omega_c^2} \right). \quad (17)$$

For standard waveguides with $2b/a = 1$, equation (17) simplifies to $\langle H_{\parallel}^2 \rangle = (H_0^2/3)(1 + \omega^2/\omega_c^2)$. Here, P_{loss} denotes the power dissipated per unit axial length of the waveguide and therefore has dimensions of power per length. This should be distinguished from P_{flow} in equation (15), which is the guided power crossing a transverse cross section and has dimensions of power.

The time-averaged Poynting theorem gives the local power balance $dP_{\text{flow}}/dz = -P_{\text{loss}}$, where P_{loss} denotes the power dissipated per unit length. For a weakly lossy uniform waveguide, we define the attenuation constant by $P_{\text{flow}}(z) = P_{\text{flow}}(0)e^{-2\alpha z}$. Thus, $dP_{\text{flow}}/dz = -2\alpha P_{\text{flow}}$, and we obtain

$$\alpha = \frac{P_{\text{loss}}}{2P_{\text{flow}}} = \frac{R_s}{Z_0 b} \frac{1 + (2b/a)(\omega_c/\omega)^2}{\sqrt{1 - (\omega_c/\omega)^2}}, \quad (18)$$

where α has units of inverse length. When α is expressed in units of m^{-1} , it can be converted to decibels per meter (dB m^{-1}) via $\alpha \rightarrow 8.686 \alpha$.

In summary, the attenuation constant α due to superconducting wall losses can be evaluated as follows. First, the normalized complex conductivity σ/σ_n is obtained from equation (3), which is then used to calculate the normalized surface resistance R_s/R_0 via equation (14). In both calculations, the input parameters are ℓ/ξ_0 , T/T_c , and $\hbar\omega/\Delta_0$ (see table 1). Finally, α is determined from equation (18), where

the explicit waveguide dimensions (a and b ; see table 2) and the intrinsic material parameter R_0 are required. Note here, $R_0 := \mu_0 \Delta_0 \lambda_0 / \hbar$ is defined from the material parameters in the pure limit at zero temperature, whereas extrinsic effects such as the mean free path and temperature are already incorporated in the normalized surface resistance $(R_s/R_0)(\ell, T, \omega)$.

2.5. TLS-induced attenuation

In the preceding sections, we evaluated the power-flow attenuation of rectangular waveguides by focusing on superconducting wall losses dominated by thermally excited quasiparticles, which are ultimately governed by the superconducting surface resistance. For practical cryogenic implementations, however, additional dissipation channels can arise from thin dielectric regions that are inevitably present at surfaces and interfaces. In particular, amorphous oxides and adsorbate layers can host tunneling TLS [17], which are known to produce a characteristic dielectric loss at low temperatures [18, 19].

We consider a thin native oxide layer (e.g. Nb_2O_5) adjacent to the waveguide walls. Within the same weak-loss, perturbative framework used for conductor loss, the TLS contribution can be obtained from the ratio of the time-averaged power dissipated in the dielectric regions, $P_{\text{loss}}^{\text{TLS}}$, to the guided power flow, P_{flow} , i.e. $\alpha_{\text{TLS}} = P_{\text{loss}}^{\text{TLS}}/(2P_{\text{flow}})$.

Denoting the complex relative permittivity of the dielectric layer by $\varepsilon_r = \varepsilon'_r - i\varepsilon''_r$ with $\varepsilon''_r = \varepsilon'_r \tan \delta_{\text{TLS}}$, the dissipated power per unit length can be written as $P_{\text{loss}}^{\text{TLS}} = (\omega/2) \int_{\text{diel}} \varepsilon_0 \varepsilon'_r \tan \delta_{\text{TLS}} |\mathbf{E}|^2 dA$, where the integral is taken over the dielectric cross-sectional area in the waveguide. Here [18, 19]

$$\tan \delta_{\text{TLS}}(\mathbf{r}) = \tan \delta_0 \frac{\tanh(\hbar\omega/2k_B T)}{\sqrt{1 + \frac{|E(\mathbf{r})|^2}{E_c^2}}} \quad (19)$$

is the TLS-induced loss tangent in the standard TLS model (see also recent studies of TLS loss in cavities [47–52]), and it depends on position through the local electric-field amplitude $E(\mathbf{r})$. The prefactor $\tan \delta_0$ is proportional to the TLS density of states P_0 and the squared dipole moment d_0^2 , while the saturation field scales as $E_c \propto 1/(d_0 \sqrt{T_1 T_2})$, where T_1 and T_2 are the TLS relaxation and dephasing times, respectively.

In our rectangular waveguides (see figure 1), for the fundamental TE_{10} mode, we have $E_x = E_z = 0$ and $E_y = E_0 \sin(\pi x/a)$, where $E_0 = i\omega\pi H_0/(\omega^2 \varepsilon_0 a)$ is the electric-field amplitude in the vacuum-filled region (see section 2.4). The walls parallel to the y axis at $x = 0$ and $x = a$ do not contribute to the loss because of the $\sin(\pi x/a)$ factor. The remaining contribution reduces to an integral over the walls parallel to the x axis, yielding

$$P_{\text{loss}}^{\text{TLS}} = \frac{\omega \varepsilon_0 a t_{\text{diel}} |E_0|^2}{2\varepsilon'_r} S(E_0) \tan \delta_0 \tanh \frac{\hbar\omega}{2k_B T}, \quad (20)$$

where

$$\begin{aligned} S(E_0) &:= \frac{2}{a} \int_0^a dx \frac{\sin^2(\pi x/a)}{\sqrt{1 + (E_0/\tilde{E}_c)^2 \sin^2(\pi x/a)}} \\ &= \frac{4}{\pi (E_0/\tilde{E}_c)^2} \left[E\left(-\frac{E_0^2}{\tilde{E}_c^2}\right) - K\left(-\frac{E_0^2}{\tilde{E}_c^2}\right) \right] \end{aligned} \quad (21)$$

with $\tilde{E}_c := \varepsilon'_r E_c$. Here t_{diel} denotes the thickness of the dielectric surface layer, which is typically of order 1 nm. K and E denote the complete elliptic integrals of the first and second kinds, respectively. Using the Taylor expansions around $E_0 = 0$, $K(-s) = (\pi/2)(1 - s/4 + 9s^2/64 + \dots)$ and $E(-s) = (\pi/2)(1 + s/4 - 3s^2/64 + \dots)$, we obtain $S(0) = 1$. At $E_0 = \tilde{E}_c$, $S \simeq 0.77$, and $S(E_0) \rightarrow 0$ as $E_0 \rightarrow \infty$. The value of E_c depends on the material and on surface processing (see, e.g. [51]).

The TLS-induced power attenuation $\alpha_{\text{TLS}} = P_{\text{loss}}^{\text{TLS}}/2P_{\text{flow}}$ is then given by

$$\alpha_{\text{TLS}} = \frac{\omega^2}{\beta c^2} \frac{t_{\text{diel}}}{\varepsilon'_r b} S(E_0) \tan \delta_0 \tanh \frac{\hbar\omega}{2k_B T}. \quad (22)$$

This result admits a simple physical interpretation. Using the group velocity $v_g = d\omega/d\beta = \beta c^2/\omega$, the prefactor can be rewritten as $\omega^2/(\beta c^2) = \omega/v_g$, which has units of inverse length, as expected for an attenuation constant. The second factor can be identified with the filling factor F . For the TE_{10} mode considered here, the electric energy stored in the dielectric layer and the total electric energy in the waveguide are $U_{\text{E,diel}} = (1/4)(\varepsilon_0/\varepsilon'_r) a t_{\text{diel}} |E_0|^2$ and $U_{\text{E}} = (1/8) \varepsilon_0 a b |E_0|^2$, respectively. Thus, $F :=$

$U_{\text{E,die}}/U_{\text{E}} = 2t_{\text{die}}/(\varepsilon_r' b)$, which quantifies the fraction of electric energy residing in the dielectric layer (e.g. $F \sim 10^{-7}$ for $t_{\text{die}} \sim 1$ nm, $\varepsilon_r' \sim 10$, and $b \sim 1$ mm).. Accordingly, α_{TLS} scales as $(\omega/\nu_g) F \tan \delta_0$, multiplied by the thermal factor $\tanh(\hbar\omega/2k_{\text{B}}T)$ and the saturation factor $S(E_0)$.

It is convenient to express the photon occupation in terms of the ac field amplitude. The photon number per unit length is $U_{\text{tot}}/\hbar\omega$, and the photon number density follows by dividing by the cross section ab . We thus obtain $n = \varepsilon_0 |E_0|^2 / 4\hbar\omega = (\omega/2\omega_c)(\mu_0 H_0^2/2)/\hbar\omega_c$. For example, for a drive amplitude $\mu_0 H_0 = 1$ μT , the corresponding electric-field amplitude is $E_0 \simeq 300$ V m^{-1} , which gives a photon number density $n \sim 10^{10}$ cm^{-3} at $f \sim f_c^{\text{WR15}} = 39.9$ GHz.

3. Examples of power-flow attenuation calculations

We now evaluate the attenuation constant for representative materials. Before doing so, we note an important point regarding the gap ratio within BCS-based modeling. In weak-coupling BCS theory, the zero-temperature gap obeys $\Delta_0 = A_{\text{BCS}} k_{\text{B}} T_c$, where $A_{\text{BCS}} = \pi/e^{\gamma_{\text{E}}} \simeq 1.76$ and $\gamma_{\text{E}} \simeq 0.577$ is the Euler constant. This relation is a good approximation for many conventional superconductors such as Al, Sn, and In. In contrast, superconductors with higher gap frequencies, including Nb, NbN, NbTiN, and Nb₃Sn, which are of practical interest for millimeter-wave and terahertz applications, often exhibit strong-coupling effects and satisfy $\Delta_0 = A k_{\text{B}} T_c$ with $A > A_{\text{BCS}}$ (for example, $A \simeq 1.9$ for Nb). Because the present theory is formulated within the weak-coupling BCS framework, using A_{BCS} may overestimate the dissipation in strong-coupling materials at a given temperature. This can be understood from the fact that dissipation is closely tied to the thermally excited quasiparticle population, which scales approximately as $n_{\text{qp}} \propto e^{-\Delta/k_{\text{B}}T} = e^{-A_{\text{BCS}} T_c/T}$. In strong-coupling superconductors, however, one should replace A_{BCS} by A , which yields a smaller quasiparticle factor and hence a lower dissipation. To account for this strong-coupling correction within a BCS-based calculation, the gap function $\Delta(T)$ obtained from the gap equation (equation (1)) is often rescaled as

$$\Delta(T) \rightarrow \frac{A}{A_{\text{BCS}}} \Delta(T), \quad (23)$$

which reproduces the weak-coupling result when $A = A_{\text{BCS}}$. For Nb and NbN, for example, this prescription gives $\Delta^{\text{Nb}}(T) = 1.08 \Delta(T)$ for $A_{\text{Nb}} = 1.9$ and $\Delta^{\text{NbN}}(T) = 1.14 \Delta(T)$ for $A_{\text{NbN}} = 2$, respectively. This prescription should be regarded as a phenomenological correction within a BCS-based calculation, rather than a self-consistent strong-coupling calculation. In the weak-coupling limit, the Eliashberg linear response reduces to the BCS response used here, whereas for strong-coupling materials, especially near the pair-breaking edge $\hbar\omega \simeq 2\Delta_0$, detailed Eliashberg self-energy effects can modify the optical conductivity. A more accurate quantitative treatment of this regime would require a full Eliashberg calculation and is beyond the scope of the present work. In the following computations for strong-coupling materials, we adopt this rescaling procedure.

3.1. NbN and Nb₃Sn

NbN is one of the most widely used superconducting materials in the superconducting-device community, with a critical temperature reaching up to $T_c \simeq 16$ – 17 K depending on film quality and stoichiometry. To evaluate the attenuation constant for niobium nitride (NbN) waveguides, we assume a strong-coupling gap ratio $A_{\text{NbN}} = 2$, a gap frequency $2\Delta_0^{\text{NbN}}/\hbar = 1.4$ THz, and a clean-limit penetration depth $\lambda_0^{\text{NbN}} = 130$ nm. These parameters correspond to the material-dependent normalization factor $R_0^{\text{NbN}} = 0.72$ Ω (see equation (13)). This allows us to convert the normalized surface resistance R_s/R_0 obtained in the previous section into the dimensional quantity R_s .

Nb₃Sn, whose gap frequency exceeds the terahertz range, is widely used in the superconducting accelerator community and has a relatively high critical temperature $T_c \simeq 18$ K. Moreover, mature techniques for producing high-quality Nb₃Sn films on the inner surfaces of superconducting radio-frequency (SRF) cavities have been established [53–56]. The material parameters routinely achievable in Nb₃Sn are comparable to the best values reported for NbN. Specifically, we assume a strong-coupling gap ratio $A_{\text{Nb3Sn}} = 2$, a gap frequency $2\Delta_0^{\text{Nb3Sn}}/\hbar = 1.46$ THz, and a clean-limit penetration depth $\lambda_0^{\text{Nb3Sn}} = 90$ nm, which yield the material-dependent normalization factor $R_0^{\text{Nb3Sn}} = 0.52$ Ω (see equation (13)).

To emphasize once again, although R_0 is defined in terms of pure-limit parameters, it serves only as a normalization factor. The calculation of R_s itself fully incorporates an arbitrary mean free path and is therefore applicable across the entire range from the clean limit to the dirty limit.

Figures 5(a) and (b) show α for NbN and Nb₃Sn rectangular waveguides at $T/T_c = 0.25$, corresponding to a temperature of order the liquid-helium temperature ($T \sim 4$ K). For clean cases (blue

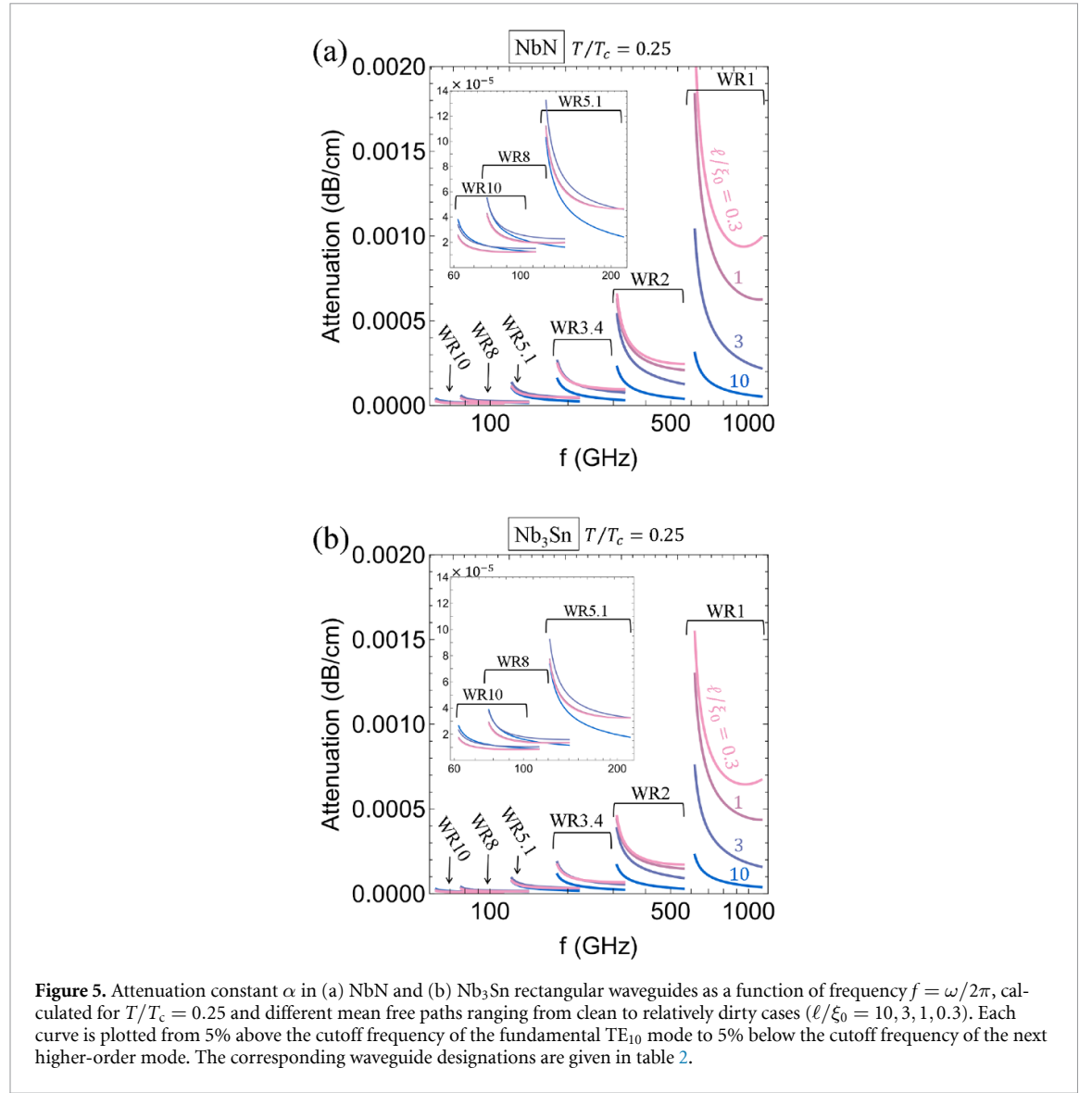


Figure 5. Attenuation constant α in (a) NbN and (b) Nb₃Sn rectangular waveguides as a function of frequency $f = \omega/2\pi$, calculated for $T/T_c = 0.25$ and different mean free paths ranging from clean to relatively dirty cases ($\ell/\xi_0 = 10, 3, 1, 0.3$). Each curve is plotted from 5% above the cutoff frequency of the fundamental TE₁₀ mode to 5% below the cutoff frequency of the next higher-order mode. The corresponding waveguide designations are given in table 2.

curves), the attenuation increases slowly as the waveguide dimension decreases, enabling highly efficient power transmission even in high-frequency waveguides such as WR1. In contrast, as the mean free path decreases, both R_s and consequently α exhibit a pronounced frequency dependence. For dirtier cases, the transmission efficiency of the waveguide is significantly reduced as the waveguide dimensions become smaller (see, for example, the pink curve for WR1). This trend can be understood by referring back to the calculation of the surface resistance shown in figures 3 and 4, where R_s/R_0 for clean material ($\ell/\xi_0 = 10$) decreases slowly with frequency below the gap frequency, except in the very low-frequency regime ($\hbar\omega \ll \Delta$). The insets in figures 5(a) and (b) provide enlarged views of the low-frequency regions, focusing on WR10, WR8, and WR5.1. In this low-frequency regime, as seen in figure 4(b) and as is well known in the accelerator community, the mean free path dependence of R_s is nonmonotonic (see e.g. [21, 44–46]). As a result, relatively dirty material (pink) is not necessarily disadvantageous as a waveguide material in the low-frequency limit ($\hbar\omega \ll \Delta$).

3.2. Niobium

In this section, we evaluate α for a Nb waveguide in the same manner as for NbN and Nb₃Sn in the previous section. Before doing so, we note an important limitation specific to Nb. The electromagnetic response of a superconductor is, in principle, nonlocal: the current at a given point is determined not only by the electromagnetic field at that point but also by the field in its vicinity. The relevant nonlocality length is of order the coherence length in the clean limit and is shortened by impurity scattering in dirtier samples. The local formulation used in this work is justified when this nonlocality length is much shorter than the length scale over which the electromagnetic field varies, namely the penetration depth λ . Thus, in the clean limit, the validity of the local approximation is

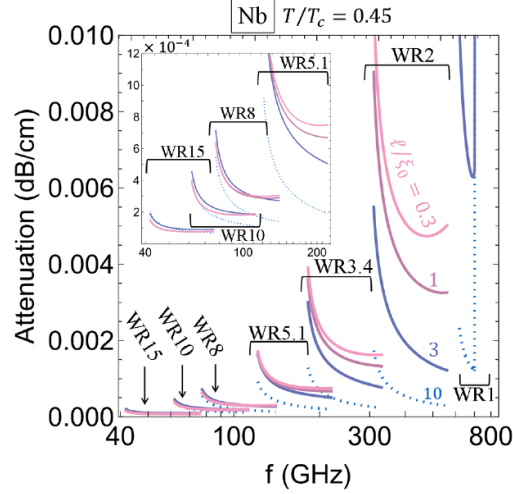


Figure 6. Attenuation constant α in Nb rectangular waveguides as a function of frequency f at $T/T_c = 0.45$, calculated for different mean free paths ranging from clean to relatively dirty cases ($\ell/\xi_0 = 10, 3, 1, 0.3$). The results for clean-limit Nb ($\ell/\xi_0 = 10$), which are strictly speaking outside the range of the present theory but still expected to be qualitatively valid, are shown as blue dotted curves for reference. Each curve is displayed from 5% above the cutoff frequency of the fundamental TE_{10} mode to 5% below the cutoff frequency of the next higher-order mode. The corresponding waveguide designations are listed in table 2.

controlled by ξ_0/λ : it is well justified for $\xi_0/\lambda \ll 1$, but it is not controlled by a small parameter when $\xi_0/\lambda \sim 1$.

This point is particularly relevant for clean-limit Nb. In the clean limit ($\ell/\xi_0 \gg 1$), Nb has a penetration depth comparable to the coherence length: $\xi_0/\lambda_0 \simeq 1$. Therefore, the dimensionless parameter controlling the omitted nonlocal correction is not small, and deviations of quantities such as R_s or Z_s from the local prediction cannot be assumed to be small on the basis of the present local theory alone. This contrasts with most other type-II superconductors, for which the penetration depth is typically much larger than the coherence length. A reliable quantitative evaluation of clean-limit Nb requires a full nonlocal calculation [44]. However, the present theory remains applicable to dirty-limit ($\ell/\xi_0 \ll 1$) and moderately dirty Nb ($\ell/\xi_0 \sim 1$), where impurity scattering reduces the effective nonlocality length and increases the penetration depth. For this reason, in addition to the dirty-limit and moderately dirty cases, we also show a clean-limit parameter set ($\ell/\xi_0 = 10$) as a reference, but it should not be interpreted as a reliable quantitative prediction.

Nb is a superconductor with $T_c \simeq 9.2$ K and is widely used as the material of choice for accelerating cavities (see, e.g. [45, 60, 61]) and high-Q resonant cavities for circuit quantum electrodynamics [47–50]. By combining high-purity Nb with a residual-resistivity ratio $RRR \gtrsim 300$ and well-established surface treatments, exceptionally high quality factors have been demonstrated: $Q > 10^{11}$ at $T \sim 1$ K [51, 52, 62–64], $Q > 10^{10}$ at millikelvin temperatures in elliptical cavities [47], and $Q > 10^9$ at millikelvin temperatures in coaxial cavities [49, 50]. The achievable accelerating field (see [61] and references therein) typically reaches an intermediate level between the lower critical field and the superheating field, which is the stability limit of the Meissner state [65–71], implying that the associated shielding currents can become comparable in magnitude to the depairing current [21, 37, 67, 68]. Owing to the high level of maturity of bulk-Nb technology, Nb is also a promising material platform for superconducting waveguides operating at frequencies below the gap scale, i.e. for $f \lesssim 700$ GHz.

We assume a strong-coupling gap ratio $A_{Nb} = 1.9$, a gap frequency $2\Delta_0^{Nb}/h = 720$ GHz, and a clean-limit zero-temperature penetration depth $\lambda_0^{Nb} = 40$ nm. These parameters correspond to the material-dependent normalization factor $R_0^{Nb} = 0.114 \Omega$ (see equation (13)).

Figure 6 shows α for Nb rectangular waveguides at $T/T_c = 0.45$, corresponding to liquid-helium temperature, while the inset provides an enlarged view of figure 6, focusing on the lower-frequency waveguides. As already seen in figure 5 for NbN and Nb₃Sn waveguides, cleaner materials yield smaller attenuation, except in the low-frequency regime, where the optimum mean free path is realized in moderately dirty materials (for details, see the corresponding discussion in the NbN and Nb₃Sn sections). The sharp increase observed for WR1 arises when the frequency approaches the gap frequency of Nb ($f \simeq 2\Delta_0^{Nb}/h$).

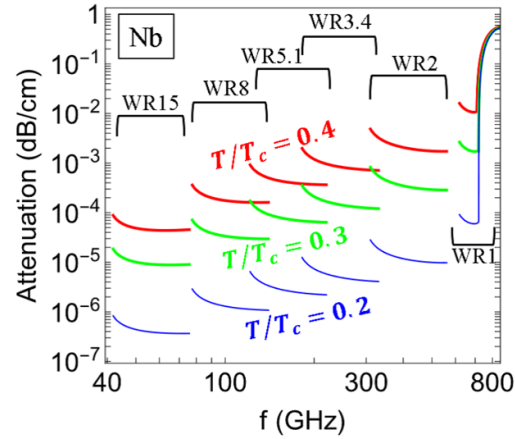


Figure 7. Attenuation constant α in Nb rectangular waveguides as a function of frequency f at $T/T_c = 0.2, 0.3, 0.4$, calculated for a moderately dirty case ($\ell/\xi_0 = 1$). Each curve is displayed from 5% above the cutoff frequency of the fundamental TE_{10} mode to 5% below the cutoff frequency of the next higher-order mode. The rapid growth of attenuation constant is seen around the gap frequency $2\Delta/h = 720$ GHz. The corresponding waveguide designations are listed in table 2.

In the above, we have calculated α for NbN, Nb₃Sn, and Nb rectangular waveguides, from which cases for any conventional type-II superconducting wall can be inferred. For a different material (e.g. NbTiN), the results scale with the magnitude of its material parameters (e.g. Δ), but the overall behavior remains qualitatively similar to that shown in figures 5 and 6.

3.3. Temperature dependence of superconducting wall loss and attenuation

It is well known that the surface resistance of an *s*-wave superconductor decreases exponentially as the temperature is lowered. Therefore, the attenuation levels shown in figures 5 and 6, which are evaluated around the liquid-helium temperature, can change significantly at different temperatures. Figure 7 shows the attenuation constant of a Nb rectangular waveguide at several temperatures for a fixed mean free path, explicitly demonstrating the exponential temperature dependence.

While the results in figure 7 are obtained from the numerical calculations based on the theoretical framework introduced above, it is also useful to provide a simple scaling estimate of $\alpha(T)$ from a reference value $\alpha(T_{\text{ref}})$. In the low-temperature regime of interest the exponential factor dominates, so a compact estimate can be obtained by retaining only the leading activation dependence. Since $\alpha(T) \propto R_s(T)$ and, for $T \lesssim 0.5T_c$, the gap is approximately temperature independent so that $R_s(T) \propto e^{-\Delta/k_B T}$, we obtain

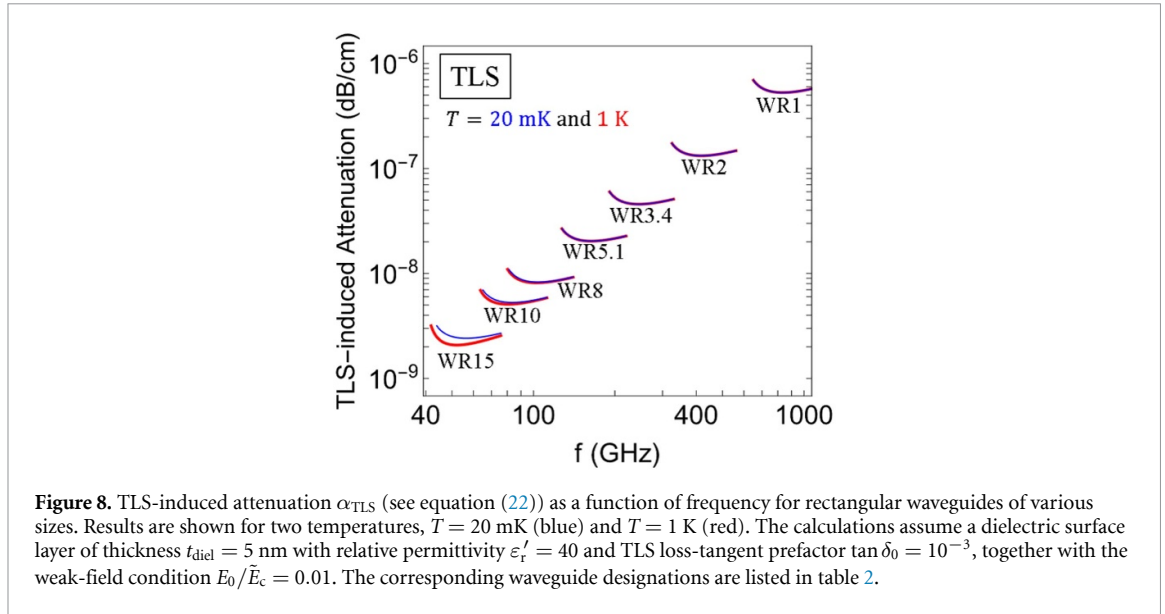
$$\frac{\alpha(T)}{\alpha(T_{\text{ref}})} = \frac{R_s(T)}{R_s(T_{\text{ref}})} \simeq \exp \left[- \left(\frac{1}{T/T_c} - \frac{1}{T_{\text{ref}}/T_c} \right) A \right]. \quad (24)$$

Here, $A = \Delta_0/(k_B T_c)$ is the gap ratio introduced in the opening paragraph of section 3.

As a consistency check, we apply the scaling relation to Nb with $A_{\text{Nb}} = 1.9$. Suppose that α is known at the reference temperature $T_{\text{ref}} = 0.4T_c$, and we estimate α at $T = 0.2T_c$ using the scaling relation. The scaling predicts $\alpha|_{T/T_c=0.2} \simeq 0.0087 \alpha|_{T/T_c=0.4}$. Using the numerical value $\alpha|_{T/T_c=0.4} = 1.75 \times 10^{-4}$ at 100 GHz (see figure 7), we obtain $\alpha|_{T/T_c=0.2} \simeq 1.5 \times 10^{-6}$. This estimate agrees reasonably well with the directly computed value $\alpha|_{T/T_c=0.2} = 1.3 \times 10^{-6}$ shown in figure 7. However, the scaling should be regarded as a simple low-temperature estimate rather than a quantitatively accurate formula over the entire frequency range. As shown in appendix C, away from the immediate gap-edge region, the deviation from the full numerical result remains within about 20% for $T/T_c = 0.3$ and about 35% for $T/T_c = 0.2$ over the subgap frequency range examined there.

More generally, once α is known at a reference temperature T_{ref} , it can be readily estimated at other temperatures via the same scaling relation. Accordingly, readers can also use this procedure to estimate α for NbN and Nb₃Sn at temperatures other than those shown in figure 5.

It should be noted that, as the temperature is lowered, the quasiparticle contribution decreases exponentially, and other loss mechanisms can become dominant. These include dissipation from trapped magnetic flux [72–76], dielectric loss associated with TLS defects at the inner surface [47–50, 52], and subgap states of unknown origin [21, 77, 78]. As a consequence, the surface dissipation often



saturates and no longer decreases below a characteristic temperature (typically $T \sim 1$ – 2 K for SRF Nb cavities).

Among these non-quasiparticle contributions, losses due to trapped magnetic flux can now be mitigated effectively by magnetic shielding and optimized cool-down procedures (see, e.g. [72–74]).

3.4. TLS contribution

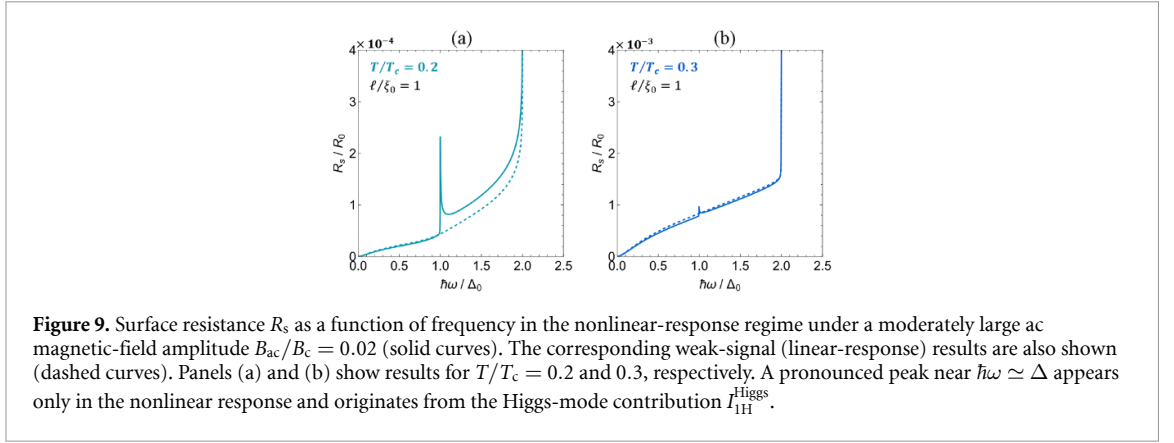
TLS-related dissipation in thin native oxide layers on superconducting surfaces remains an active area of research and a persistent challenge. Although several approaches have shown promising results [47, 50, 52], TLS loss is still widely regarded as one of the major limitations in superconducting resonator performance. In this section, we provide an order-of-magnitude estimate of the TLS-induced attenuation α_{TLS} in rectangular waveguides using parameter values of native oxide layers inferred from Nb cavity experiments.

Figure 8 shows α_{TLS} as a function of frequency obtained from equation (22) for several waveguide sizes and for two temperatures, $T = 20$ mK (blue) and $T = 1$ K (red). We assume $t_{\text{diel}} = 5$ nm, $\epsilon'_r = 40$, $\tan \delta_0 = 10^{-3}$ [50], and a weak-field condition $E_0/\tilde{E}_c = 0.01$.

Combining these results with the scaling relation in equation (24), we can estimate the temperature below which TLS loss becomes comparable to, and eventually exceeds, the quasiparticle-limited attenuation. For a 100 GHz signal transmitted through a WR10 waveguide, we find $\alpha_{\text{TLS}} \simeq 7 \times 10^{-9}$, which is comparable to the Nb attenuation α_{Nb} at $T/T_c \simeq 0.13$. Likewise, for a 300 GHz signal transmitted through a WR3.4 waveguide, we obtain $\alpha_{\text{TLS}} \simeq 7 \times 10^{-8}$, comparable to α_{Nb} at $T/T_c \simeq 0.14$. These estimates suggest that TLS-induced loss can dominate at sufficiently low temperatures, e.g. for $T/T_c \lesssim 0.1$, in which case the total attenuation is well approximated by $\alpha_{\text{tot}} \simeq \alpha_{\text{TLS}}$ with α_{TLS} given by equation (22). By contrast, for $T/T_c \gtrsim 0.2$ the quasiparticle contribution is expected to dominate, and thus $\alpha_{\text{tot}} \simeq \alpha$ as evaluated in figures 5–7.

It should be noted that these values depend on $\tan \delta_0$, which in the standard TLS model scales as $\tan \delta_0 \propto P_0 d_0^2 / \epsilon'_r$, where P_0 is the TLS density of states and d_0 is the TLS electric dipole moment. The value $\tan \delta_0 = 10^{-3}$ used above is representative of native oxide layers on the best Nb cavities processed with accelerator-like surface treatments [50]. In general, however, $\tan \delta_0$ can vary substantially with the material and with chemical and thermal processing, and it may change by orders of magnitude for Nb under different treatments as well as for NbN, Nb₃Sn, and other superconductors.

Equation (22) shows the simple scaling of the TLS contribution: for fixed geometry and field amplitude, and when the TLS thermal factor is nearly unchanged, $\alpha_{\text{TLS}} \propto t_{\text{diel}} \tan \delta_0$. Thus, if $\tan \delta_0$ is larger by a factor η , the TLS-induced attenuation increases by the same factor. The crossover temperature at which TLS loss becomes comparable to quasiparticle loss shifts upward only logarithmically, since the quasiparticle contribution scales approximately as $\exp(-AT_c/T)$. If the original crossover temperature is T_0 , the shifted crossover temperature is roughly given by $T_\eta \simeq T_0 / [1 - (T_0/T_c)(\ln \eta)/A]$.



4. Nonlinear response and Higgs mode in rectangular waveguides

4.1. Nonlinear correction to the surface resistance

As the guided power increases, linear response no longer provides an adequate description of dissipation in the waveguide walls. In this regime, the complex conductivity acquires an explicit dependence on the driving-field amplitude (see, e.g. [79, 80] for detailed experimental studies). Recently, within the Keldysh–Usadel framework for nonequilibrium superconductivity in disordered systems, the current response was computed up to order $\mathcal{O}(E^3)$, yielding a nonlinear correction $\delta\sigma_1$ to the dissipative component σ_1 [23]. The theory [23] predicts that $\delta\sigma_1$ exhibits a peak attributed to the Higgs-mode at $\hbar\omega = \Delta$ [23, 33]. In this section, focusing on the dirty limit, we investigate how these nonlinear effects modify the power-flow attenuation in rectangular waveguides.

According to the nonlinear response theory [23], the amplitude-dependent correction to the dissipative conductivity is given by

$$\frac{\delta\sigma_1}{\sigma_n} = u(T, \omega) \left(\frac{q_0}{q_\xi} \right)^2, \quad (25)$$

$$u(T, \omega) = \frac{2\sqrt{\pi}}{\hbar\omega/\Delta_0} \text{Re} \left(I_{IH}^{qqq} + I_{IH}^{\text{Higgs}} + I_{IH}^{\text{Eliash}} \right). \quad (26)$$

Here, q_0 denotes the amplitude of the superfluid momentum, and q_ξ is the inverse dirty-limit coherence length. The term I_{IH}^{qqq} arises from the direct nonlinear photon contribution. The terms I_{IH}^{Higgs} and I_{IH}^{Eliash} represent contributions mediated by nonequilibrium variations of the pair potential, originating from the Higgs mode and the Eliashberg effect, respectively. Their explicit forms are lengthy and are therefore omitted here; they can be found in equation (84) of [23].

This amplitude-dependent correction yields the surface resistance [23]

$$R_s(B_{ac}) = R_s(0) + \delta R_s(B_{ac}), \quad (27)$$

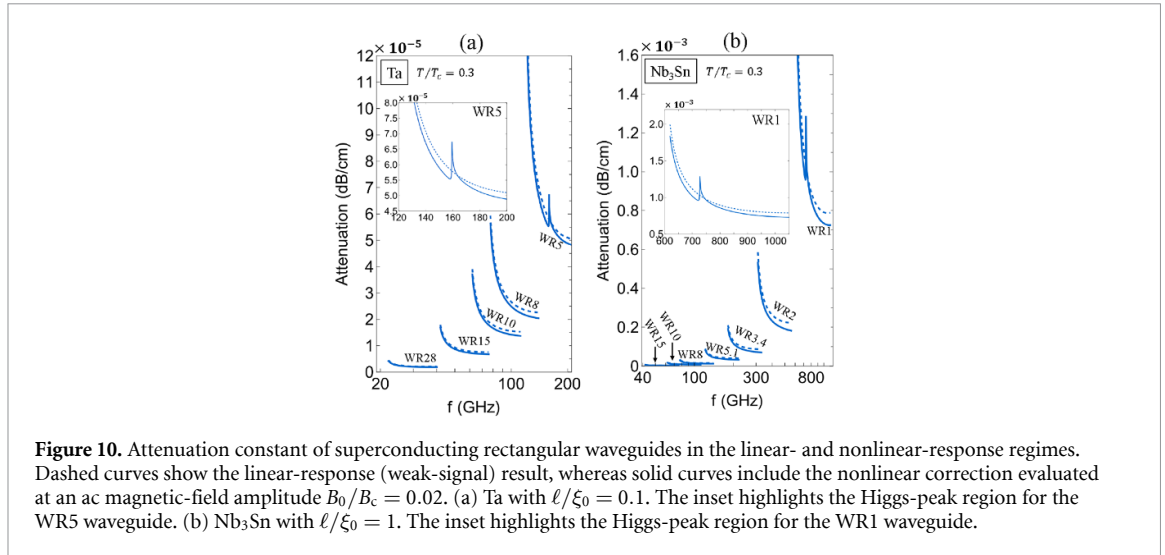
$$\delta R_s(B_{ac}) = \frac{1}{2} \mu_0^2 \omega^2 \lambda^3 \sigma_n \frac{u(T, \omega)}{2\pi} \left(\frac{B_{ac}}{B_c} \right)^2, \quad (28)$$

for $\sigma_1/\sigma_2 \ll 1$. Here, $B_c = \sqrt{\mu_0 N_0} \Delta_0$ is the zero-temperature thermodynamic critical field, and B_{ac} is the amplitude of the ac magnetic field. For our rectangular waveguide, we take $B_{ac} = \mu_0 \sqrt{\langle H_{\parallel}^2 \rangle}$, where $\langle H_{\parallel}^2 \rangle$ is given by equation (17). Normalizing by R_0 as before, we obtain

$$\frac{R_s}{R_0} = \frac{1}{2\pi \sqrt{\ell/\xi_0}} \left(\frac{\hbar\omega}{\Delta_0} \right)^2 \left(\frac{\lambda}{\lambda_{0, \text{dirty}}} \right)^3 \times \left[\frac{\sigma_1}{\sigma_n} + \frac{u(T, \omega)}{2\pi} \left(\frac{B_{ac}}{B_c} \right)^2 \right], \quad (29)$$

where u is given by equation (26), and the explicit forms of I_{IH}^{qqq} , I_{IH}^{Higgs} , and I_{IH}^{Eliash} can be found in equation (84) of [23].

Figure 9 shows the surface resistance R_s in the linear-response regime (i.e. the weak-signal limit) and in the nonlinear-response regime for a moderately large ac magnetic-field amplitude, $B_{ac}/B_c = 0.02$. The dashed and solid curves correspond to the linear and nonlinear responses, respectively. The peak at $\hbar\omega \simeq \Delta$ originates from the nonlinear Higgs-mode contribution I_{IH}^{Higgs} , which has recently been proposed as a



distinctive fingerprint of the Higgs mode that has been overlooked in conventional analyses. As seen in figures 9(a) and (b) for $T/T_c = 0.2$ and 0.3 , respectively, the Higgs-peak contrast relative to the linear-response value becomes more pronounced as the temperature is lowered.

4.2. Nonlinear attenuation

The nonlinear attenuation can be obtained by substituting the nonlinear surface resistance (equation (29)) into the attenuation formula (equation (18)). In this section, we neglect TLS-induced attenuation, whose relative contribution decreases with increasing temperature T and also with increasing ac amplitude owing to TLS saturation.

The nonlinear correction affects the attenuation in two distinct ways. First, it produces a smooth amplitude-dependent shift of the attenuation constant. Second, it gives rise to a peak at the characteristic frequency associated with the Higgs-mode resonance. It should be emphasized that the position of this Higgs-related peak is determined by the superconducting material through the condition $\hbar\omega \simeq \Delta(T)$, as shown in figure 9, and is therefore not a property of the waveguide geometry. The waveguide geometry enters only through the allowed single-mode frequency window and through the geometrical factor in equation (18). Thus, a Higgs-related peak in the attenuation constant can be observed only when the material-dependent frequency $f \simeq \Delta(T)/h$ lies within the operating band of the chosen waveguide.

Figure 10 compares the attenuation constant in the weak-signal (linear-response) limit with that in the nonlinear-response regime for a moderately large ac magnetic-field amplitude $\mu_0 H_0/B_c = 0.02$. Note that this H_0 defines the effective ac magnetic-field amplitude B_{ac} via $B_{ac} = \mu_0 \sqrt{\langle H_{\parallel}^2 \rangle}$, with $\langle H_{\parallel}^2 \rangle$ given by equation (17). The dashed and solid curves correspond to the linear and nonlinear responses, respectively.

Figure 10(a) shows the result for tantalum (Ta). The material parameters are as follows: the critical temperature is $T_c \simeq 4.4$ K, the gap ratio is close to the weak-coupling value $A \simeq A_{BCS}$, the gap frequency is $2\Delta/h = 320$ GHz, and the zero-temperature clean-limit London penetration depth is $\lambda_0 = 22$ nm, which yields the normalization factor $R_0 = 0.028 \Omega$ (see equation (13)). In the calculation, the mean free path is set to $\ell/\xi_0 = 0.1$, for which we have $\lambda \simeq \xi_0 \simeq 80$ nm. The material gap gives the Higgs-resonance frequency $f_H \simeq \Delta/h \simeq 160$ GHz, which falls within the TE_{10} window of the WR5.1 waveguide. Correspondingly, a pronounced Higgs-related feature appears at this frequency in the WR5.1 attenuation curve (see the inset).

Figure 10(b) shows the result for Nb₃Sn with $\ell/\xi_0 = 1$. For this material, the expected Higgs-resonance frequency is $f_H \simeq \Delta/h \simeq 730$ GHz, which lies within the TE_{10} window of the WR1 waveguide. Again, the attenuation curve exhibits a Higgs-related peak at this frequency in the WR1 waveguide (see the inset).

For Ta, the Higgs peak appears in a more fabrication-friendly waveguide size, which corresponds to a lower frequency. However, the attenuation at such low frequencies is small, so a sensitive measurement is required to resolve the Higgs peak. For Nb₃Sn, in contrast, the Higgs peak occurs in a more fabrication-challenging smaller waveguide (see table 2). Because the attenuation is larger at higher frequencies, the required measurement sensitivity is less stringent than in the Ta case.

It should be noted that changing the waveguide geometry does not change the material resonance itself. For a fixed material and a fixed local ac surface-field amplitude, the attenuation near the Higgs resonance can be written schematically as $\alpha(\omega) = G(a, b, \omega)R_s(\omega)$, where $G(a, b, \omega)$ is the waveguide geometrical factor in equation (18). If $G(a, b, \omega)$ varies only weakly over the narrow frequency region of the Higgs feature, both the peak and the nearby background are multiplied by nearly the same factor. Therefore, the relative Higgs-peak contrast is mainly determined by the nonlinear surface resistance, not by the waveguide geometry. The role of the waveguide geometry is instead to determine whether $f_H \simeq \Delta/h$ lies inside the TE_{10} window and to set the absolute scale of the attenuation.

For Nb_3Sn , the Higgs-resonance frequency $f_H \simeq 730$ GHz lies in the WR1 window used in figure 10. A slightly larger waveguide, such as a WR1.5-type geometry, could also include this frequency in its TE_{10} window and would be more favorable from the fabrication viewpoint. However, in such a geometry the resonance frequency would lie closer to the next-mode cutoff, reducing the single-mode margin and making the result more sensitive to fabrication tolerances and higher-order-mode effects. Thus, changing the waveguide size mainly introduces a trade-off among fabrication feasibility, single-mode margin, and the absolute attenuation level, rather than enhancing the intrinsic Higgs-peak contrast.

Although lowering the temperature enhances the Higgs-peak contrast relative to the linear-response attenuation $\alpha \propto R_s$, as seen in figure 9, it simultaneously suppresses the absolute magnitude of α , thereby placing more stringent demands on measurement sensitivity.

A straightforward way to enhance the Higgs-mode signal is to increase the ac drive amplitude (see equation (29)). In the present calculation we set $B_0/B_c \sim 10^{-2}$ (corresponding to $n \sim 10^{16} \text{ cm}^{-3}$ for WR15). Substantially larger amplitudes, up to $B_0 \sim B_c^{Nb} = 200 \text{ mT}$ (corresponding to $n \sim 10^{20} \text{ cm}^{-3}$ for WR15), are expected to significantly enhance the Higgs peak; however, treating this regime would require a nonperturbative analysis beyond the perturbative approach of [23].

5. Discussion and conclusion

In section 2 we developed a framework for calculating the power-flow attenuation α in superconducting rectangular waveguides (equations (3), (14) and (18)), applicable to arbitrary electronic mean free paths, based on the microscopic theory of superconductivity. The analytical formula of TLS-induced attenuation α_{TLS} was also derived based on the standard TLS model (equation (22)).

In section 3, we performed explicit numerical evaluations of α for representative materials (NbN , Nb_3Sn , Nb , and TLS) across standard rectangular-waveguide sizes from WR15 to WR1. A comprehensive comparison between theory and experiment over most of the parameter space remains a task for future work. One notable exception is the WR10 Nb waveguide fabricated by Nakajima *et al* [14], for which an attenuation of $5 \times 10^{-4} \text{ dB cm}^{-1}$ was reported for WR10 at $T = 4 - 5 \text{ K}$, comparable in magnitude to our results in figure 6(b). At the same time, a quantitatively reliable prediction for pure bulk Nb ultimately requires combining nonlocal electrodynamics with the rectangular-waveguide geometry. As discussed in section 3.2, clean-limit Nb has $\xi_0/\lambda_0 \simeq 1$, so the local approximation is not controlled by a small parameter. A fully nonlocal calculation of the attenuation constant for clean-limit Nb waveguides is therefore an important subject for future work.

Our results suggest that the design strategy for achieving low attenuation is relatively straightforward, at least in the high-frequency regime $f \gtrsim 0.5\Delta/h$. In this regime, low loss requires clean materials with $\ell \gtrsim \xi_0$ (see figures 5 and 6). In the case of Nb , high-purity material with $RRR \gtrsim 300$ is commercially available, largely as a result of sustained collaboration between the accelerator community and industry in the 1980 s and 1990 s [61, 81]. Combined with established SRF surface-treatment protocols, Nb rectangular waveguides are therefore expected to achieve very low attenuation below the Nb gap frequency (e.g. $2\Delta^{Nb}/h \simeq 700 \text{ GHz}$). For operation deeper into the terahertz regime, however, materials with larger gaps than Nb are required, such as NbN , Nb_3Sn , and related superconductors. In addition, these materials must typically be implemented as coatings on the waveguide walls, making film-growth or coating technology compatible with waveguide geometries such as figure 1 (or split-block implementations) essential. In this respect, vapor-diffusion growth of Nb_3Sn [53] developed for accelerator cavities [53–56] may provide promising routes. MgB_2 coatings [57–59], which involve a two-gap superconductor beyond the scope of the present single-gap framework, may provide an additional promising route. In particular, as frequencies approach the THz regime the smaller π -band gap is expected to set the relevant loss scale (roughly $2\Delta_\pi/h \simeq 900 \text{ GHz}$), making MgB_2 especially attractive for the sub-THz to near-THz range. To treat MgB_2 quantitatively, the present single-gap framework

must be generalized to band-dependent Green functions and gaps, e.g. for the σ and π bands. The self-consistency equation should then be replaced by coupled gap equations with a pairing matrix $\lambda_{\alpha\beta}$, and the conductivity should be obtained by summing the band-resolved responses with appropriate density-of-states and Fermi-velocity weights. Such a multigap extension is beyond the scope of the present work.

In the deeply submillimeter regime, the geometrical accuracy of the waveguide also becomes important. Although the present theory gives the surface-resistance-limited attenuation for an ideal rectangular geometry, surface roughness, dimensional errors, and split-block steps in small-aperture waveguides such as WR2 and WR1 can perturb the TE_{10} field distribution and cause additional scattering, mode conversion, or local field enhancement. Such geometry-dependent corrections are beyond the scope of the present work and should be evaluated by full-wave simulations or dedicated measurements of the actual fabricated structures.

The TLS-induced attenuation α_{TLS} (equation (22)), evaluated using a parameter set representative of native Nb oxides on SRF-grade Nb, is shown in figure 8. For the assumed parameters, the resulting attenuation is extremely small: $\alpha_{\text{TLS}} \sim 10^{-9} \text{ dB cm}^{-1}$ for WR15, and it remains below $\sim 10^{-6} \text{ dB cm}^{-1}$ even for WR1. At higher drive powers, the TLS saturation factor $S(E_0)$ further suppresses dissipation. Consequently, for Nb at $T/T_c \gtrsim 0.2$ (i.e. $T \gtrsim 2 \text{ K}$), the TLS contribution is negligible compared with the quasiparticle-limited attenuation (cf figures 7 and 8).

For higher-gap materials relevant to terahertz operation, such as Nb_3Sn , the TLS parameters of the surface dielectric layers have not yet been established in comparable detail. Nevertheless, cavity measurements indicate that the surface resistance $R_s(T \ll T_c)$ of Nb_3Sn is only modestly larger than that of Nb and remains of the same order of magnitude [53, 54]. Although the residual surface resistance can have multiple microscopic origins, this suggests that, with appropriate surface processing, TLS-related loss in Nb_3Sn waveguides may be reducible to a level comparable to that estimated in figure 8.

In section 4, we investigated nonlinear attenuation using a recently developed nonlinear-response theory within the Keldysh–Usadel framework of nonequilibrium superconductivity [23]. As shown in figure 10, the Higgs-mode contribution produces a pronounced peak in α near $f \simeq \Delta/h$, corresponding to $f \simeq 160 \text{ GHz}$ for Ta, 360 GHz for Nb, and 700 GHz for NbN and Nb_3Sn . This Higgs-mode signature originates from the Kerr-type nonlinearity of the dissipative conductivity σ_1 [23, 33] and is distinct from the Higgs signatures inferred from third-harmonic generation [23–33], which are often regarded as a canonical hallmark of the Higgs mode in disordered superconductors. The Higgs peak associated with this Kerr nonlinearity has been largely overlooked in earlier analyses and has only recently been highlighted in [23, 33].

For Ta, the Higgs-resonance frequency lies near $f_H \simeq 160 \text{ GHz}$, where the absolute attenuation is extremely small. Therefore, resolving the nonlinear change in attenuation would be challenging even in a differential power-dependent measurement. A more realistic route would be to use a resonator-enhanced waveguide structure, as in the Nb waveguide experiment of Nakajima *et al*, so that the small distributed loss is converted into a measurable change in resonance linewidth or quality factor. For higher-gap materials such as NbN and Nb_3Sn , the Higgs-resonance frequency is shifted to $f_H \sim 700 \text{ GHz}$, where the attenuation is larger and the peak should be easier to resolve in principle, provided that a sufficiently low-loss WR1-scale waveguide or waveguide resonator can be fabricated. A realistic implementation would be a split-block rectangular waveguide whose inner walls are coated with a superconducting film thicker than several penetration depths, for example by conformal coating methods such as ALD-type processes for nitrides or by vapor-diffusion growth for Nb_3Sn .

In figure 10, we considered a moderately strong ac field ($B_0/B_c = 0.02$); the peak height is expected to increase with increasing drive amplitude. Extending the theory to substantially stronger fields will require a nonperturbative solution of the Keldysh–Usadel equations that consistently accounts for field-induced modifications of the quasiparticle spectrum, in the spirit of [82, 83]. Such a treatment remains an important direction for future work.

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Data availability statement

The data that support the findings of this study will be openly available following an embargo at the following URL/DOI: <https://doi.org/10.5281/zenodo.20020106> [84]. Data will be available from 31 December 2026.

Appendix A. Low-frequency formula for σ_2

The imaginary part of the complex conductivity, σ_2 , can be calculated at all temperatures and frequencies from the general expression (equation (3)). There also exists a widely used approximation, $\sigma_2 = 1/(\mu_0\omega\lambda^2)$, which is obtained from equation (3) by expanding in the small parameter ω (see equation (10)). This low-frequency formula is frequently employed in studies of superconducting devices, although its limited range of validity is not always well recognized across the superconducting device communities. It is therefore useful to explicitly compare σ_2 obtained from the low-frequency approximation with that obtained from the general expression.

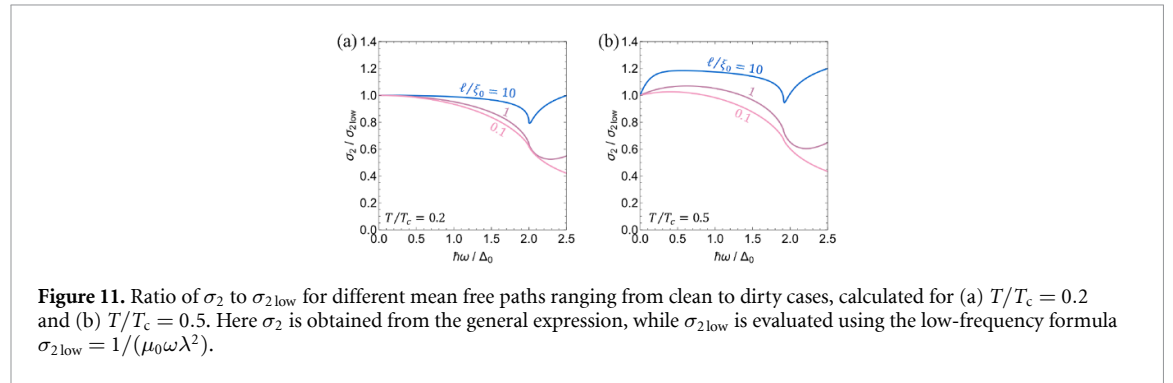


Figure 11 shows the ratio $\sigma_2/\sigma_{2\text{low}}$. Here σ_2 is obtained from the general expression, while $\sigma_{2\text{low}}$ is evaluated using the low-frequency formula $\sigma_{2\text{low}} = 1/(\mu_0\omega\lambda^2)$. For $T/T_c = 0.2$ in the clean case, $\sigma_{2\text{low}}$ agrees well with σ_2 up to frequencies close to 2Δ , whereas at higher temperatures or impurity levels the difference can become non-negligible. In particular, when calculations near the gap frequency (2Δ) are required, as in the present study, the general expression should be used.

Appendix B. σ_1/σ_2 is not necessarily negligibly small

The assumption $\sigma_1/\sigma_2 \ll 1$ is often employed in studies of superconducting devices, for example when deriving the approximate surface-resistance formula $R_s = (1/2)\mu_0^2\omega^2\lambda^3\sigma_1$. Although this assumption is not universally valid, explicit comparisons of σ_1/σ_2 over a wide range of parameters have rarely been presented. For this reason, we provide illustrative results in this appendix, which may be useful for readers as a reference.

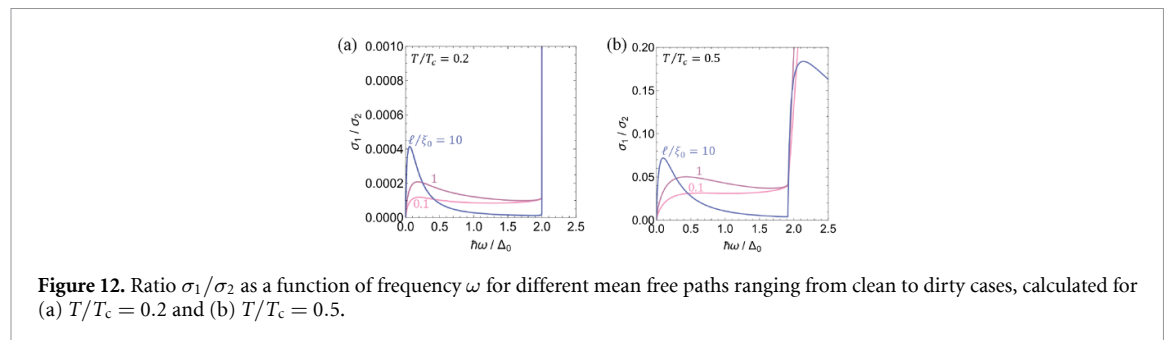
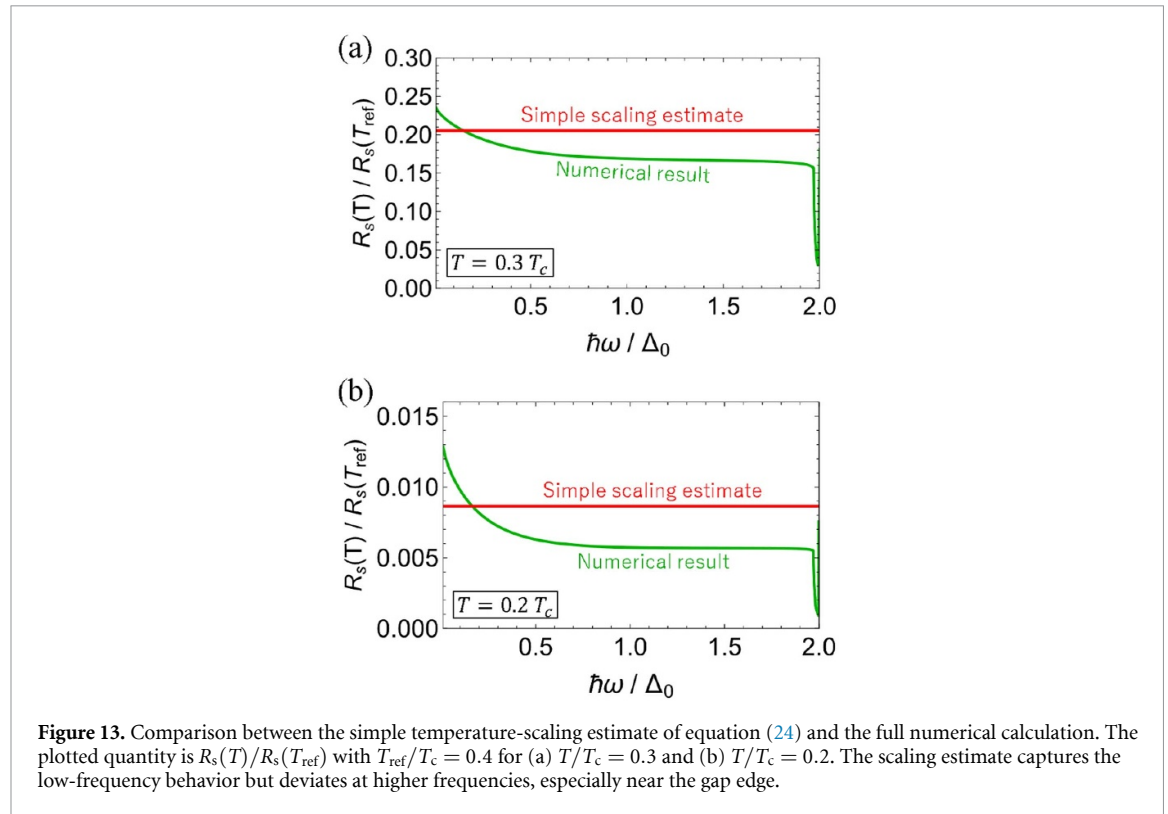


Figure 12 shows the ratio $(\sigma_1/\sigma_2)(\omega)$ for different mean free paths and temperatures. At low temperature ($T/T_c = 0.2$), σ_1/σ_2 remains negligibly small for all mean free paths as long as the frequency is below the gap frequency $\hbar\omega = 2\Delta$. As the temperature increases, however, the ratio σ_1/σ_2 acquires non-negligible values even at frequencies well below the gap frequency.

Appendix C. Validity of the simple temperature-scaling estimate

In section 3.2, we introduced a simple temperature-scaling estimate for the attenuation constant based on the leading activation factor of thermally excited quasiparticles. Since the waveguide geometrical factor in equation (18) is independent of temperature at fixed frequency and fixed geometry, the same scaling can be tested directly at the level of the surface resistance. For $T \lesssim 0.5T_c$, where the gap is approximately temperature independent, the estimate reads $\alpha(T)/\alpha(T_{\text{ref}}) = R_s(T)/R_s(T_{\text{ref}}) \simeq \exp[-A(T_c/T - T_c/T_{\text{ref}})]$.

Figure 13 compares this scaling estimate with the full numerical calculation for $\ell/\xi_0 = 1$, using $A = 1.9$ and $T_{\text{ref}}/T_c = 0.4$. The scaling reproduces the low-frequency behavior reasonably well, especially as an order-of-magnitude estimate. However, deviations become larger as the frequency approaches the gap edge, where the detailed frequency dependence of the complex conductivity and the spectral structure near the pair-breaking threshold become important. Thus, the scaling formula should be understood as a low-temperature, low-frequency estimate based on the leading activation factor, not as a replacement for the full numerical calculation over the entire frequency range.



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